■■ DataSentinel — Pipeline Quality Report

Pipeline: **HR Employee Directory Quality Pipeline** | Target Source: hr_employees | Run ID: run-e8ec37e1 | Date: 2026-10-30 09:33:43 UTC

Quality Pass Rate	Total Records	Failed Rules	Anomalies	Schema Drift
16.7%	20	4	4	STABLE

1. Quality Rules Validation Breakdown

Status	Rule ID	Type	Column	Failures	Fail %	Details
PASSED	RULE-HR-001	null_check	employee_id	0/20	0.0%	Column 'employee_id' has zero null values as required.
FAILED	RULE-HR-002	unique_check	employee_id	2/20	10.0%	Column 'employee_id' uniqueness rate 95.00% is below required 100.00%. Found 2 n...
FAILED	RULE-HR-003	null_check	work_email	1/20	5.0%	Column 'work_email' has 1 null values (5.00%); expected 0.
WARNING	RULE-HR-004	regex	work_email	1/20	5.0%	Column 'work_email' has 1 values that do not match pattern '^[a-zA-Z0-9._%+-]+@c...
FAILED	RULE-HR-005	range	years_experience	1/20	5.0%	Column 'years_experience' has 1 values outside range (>= 0.0 and <= 50.0). Sampl...
FAILED	RULE-HR-006	foreign_key	department	1/20	5.0%	Column 'department' has 1 orphaned/invalid foreign keys not found in set of 7 re...

2. Statistical Anomaly Detection (Z-Score & IQR)

Column	Method	Outliers	Outlier Rate	Distribution Metrics	Top Anomalous Samples
salary	ZSCORE	1	5.00%	μ=194150.0, σ=391153.6614	1850000.0 (conf=0.9111)
salary	IQR	1	5.00%	Q1=78750.0, Q3=136250.0	1850000.0 (conf=1.0)
years_experience	ZSCORE	1	5.00%	μ=6.75, σ=5.3888	25.0 (conf=0.6289)
years_experience	IQR	1	5.00%	Q1=3.75, Q3=8.25	25.0 (conf=1.0)